

- 1** An operation is to be performed on the ASCII values of each character in a word. The operation is described by the pseudo-code below.

```

01 c = 0
02 INPUT i
03 WHILE i <> -1 DO
04     INPUT n
05     IF n > i THEN
06         c = c + 1
07     ENDIF
08     SWAP (n, i)
09 ENDWHILE
10 OUTPUT c

```

SWAP is a predefined procedure that swaps the contents of its two parameters.

- (a) Draw a flowchart to represent the operation. SWAP should be shown as a single process in your flowchart. [4]
- (b) The ASCII codes for the letters in the word 'HELLO' are input into the program and are terminated with -1. The ASCII codes are 72, 69, 76, 76, 79.
Create and complete a trace table for this operation. Use the following headings:

c	i	n	OUTPUT
---	---	---	--------

[4]

- (c) Outline the purpose of this operation. [1]
- (d) The denary value representing the character 'O' in ASCII is 79.
- (i) Convert the denary value 79 to hexadecimal. [1]
 - (ii) Convert the denary value 79 to binary. [1]
 - (iii) Explain **one** limitation of using ASCII as a method for encoding characters. [2]

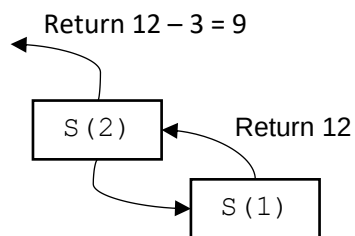
2 The recursive function S has one parameter, n .

```

01 FUNCTION S (n : INTEGER) RETURNS INTEGER
02     IF n = 1 THEN
03         RETURN 12
04     ELSE
05         RETURN S (n-1) - 3
06     ENDIF
07 ENDFUNCTION

```

(a) An example of a trace tree diagram showing $S(2)$ is shown as follows:



Use the above example to create a trace tree diagram for the recursive function call $S(5)$.

- (b) Outline the features of function $S()$ that show it is a recursive function. [4]
- (c) Explain how a stack will be used when $S(2)$ is called. [3]
- (d) Explain what would happen to the recursion and execution of the function if the call $S(0)$ is made. [4]
- (e) Rewrite the function $S()$ in pseudo-code without using recursion. [2]

- 3 A programmer is writing a simulation to model the behaviour of bees. Bees are categorised as queens, workers or drones.

For each bee, the following items of data are stored:

- a unique identification number
- status (active or inactive)
- gender (drones are male and queens and workers are female)
- energy (initialised to a random number between 50 and 100 inclusive).

For queens, the additional data item stored is:

- number of eggs per day (an integer between 2000 and 3000 inclusive).

For workers, the additional data items stored are:

- milligrams of nectar gathered per day (an integer between 0 and 20 inclusive)
- milligrams of pollen gathered per day (an integer between 0 and 20 inclusive).

For drones, there are no additional data items stored.

The simulation runs one cycle at a time. During each cycle, the amount of energy a bee has is calculated:

- a queen bee's energy decreases by 2
- a worker bee's energy will be modified by adding the amount of nectar gathered per day and then subtracting the amount of pollen gathered per day
- a drone bee's energy decreases by 1

If the energy a bee has is less than or equal to 0, the bee's status is set to inactive.

Object-oriented programming will be used to model the simulation.

(a) Draw a class diagram that shows the following for the simulation described above:

- the superclass
- any subclasses
- inheritance
- attributes
- appropriate methods.

[10]

(b) Define what is meant by instantiation.

[1]

(c)

(i) Describe encapsulation using an example from the simulation.

[2]

(ii) Describe inheritance using an example from the simulation.

[2]

(iii) Describe polymorphism using an example from the simulation.

[2]

(d) Copy and complete the table below to give an appropriate example of each type of test data that can be used to test the input of milligrams of nectar gathered per day for a worker bee.

Type of test	Test data
Normal	
Abnormal	
Extreme	

[3]

4 An estate agency helps customers to buy property.

The agency employs people who are referred to as agents. The agency records the following data for each agent:

- agent reference number
- name
- salary.

A customer registers with the agency. The agency records the following data for each customer:

- customer reference number
- name
- mobile phone number
- email address.

The agency records the following data for each property it has on sale:

- property reference code
- address
- price in dollars
- number of bedrooms.

Viewings are conducted by agents who work for the agency. The agency records the following details for each viewing:

- the property being viewed
- the customer viewing the property
- the agent conducting the viewing
- date of viewing
- time slot (am or pm).

Customers can make more than one viewing.

The agency will only ever arrange one viewing for a given property on a given date and time slot.

(a) The estate agency wants to use a relational database to store the data for the system.

(i) A database needs a number of tables to store the data.

Draw an entity-relationship (ER) diagram to show the tables in third normal form (3NF) and the relationship(s) between them. [4]

(ii) A table description can be expressed as:

TableName (Attribute1, Attribute2, Attribute3, ...)

The primary key is indicated by underlining one or more attributes. Foreign keys are indicated by using a dashed underline.

Write table descriptions for the tables you identified in part **(a)(i)**, using the information given. [7]

(iii) Write an SQL query to create the property table from your table description in part **(a)(ii)**. [4]

(b) The estate agency will need to back up and archive data.

(i) Explain the need to back up data. [2]

(ii) Explain the need to archive data. [2]

- 5 A binary search tree holds the names of capital cities in the array `bst`. Each element of the array contains three values. `leftPtr` and `rightPtr` are integers and `dataValue` is a string.

<code>leftPtr</code>	<code>dataValue</code>	<code>rightPtr</code>
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The root of the binary search tree is stored in an integer variable, `rootPtr`.

The unused elements of the array are in a free space list that starts from `freePtr`.

The provided names of capital cities are guaranteed to be unique.

The contents of the array `bst` is shown below. `-1` represents the null pointer.

Index	leftPtr	dataValue	rightPtr			
0	5	London	1	<table><tr><th>rootPtr</th></tr><tr><td>0</td></tr></table>	rootPtr	0
rootPtr						
0						
1	2	Paris	3			
2	-1	New York	-1			
3	-1	Rome	4			
4	-1	Tokyo	-1			
5	-1	Beijing	-1			
6	7		-1	<table><tr><th>freePtr</th></tr><tr><td>6</td></tr></table>	freePtr	6
freePtr						
6						
7	8		-1			
8	9		-1			
9	-1		-1			

An individual value in the array can be accessed in the following format: `bst[0].leftPtr`

- (a)
- Draw the binary search tree represented by the array `bst`. [2]
 - Give **two** properties of a binary search tree. [2]
 - The city of Madrid is to be inserted into the binary search tree. Identify the changes that will be made to any pointers and the contents of the array `bst`. [3]
- (b) `bst` has been implemented as a static data structure of 10 nodes. State **two** problems that can arise from using a static data structure. [2]
- (c) A reverse in-order traversal of a binary tree can be written recursively as:
- follow the right pointer and recursively repeat from step 1
 - output the current node
 - follow the left pointer and recursively repeat from step 1
- Write a recursive pseudo-code procedure `reverseInOrder()` that takes the value of `rootPtr` and outputs the cities in `bst` using a reverse in-order traversal. [6]
 - Give the output of performing the reverse in-order traversal on `bst` from part (a). [1]

- 6** A company has multiple offices and a head office. Alice and Bob work at different offices and can both access the company email system and intranet that are hosted at the company head office. The company has a client-server architecture.
- (a) Describe the term client-server architecture. [2]
 - (b) Describe the term intranet. [2]
 - (c) The company email server uses the SMTP protocol and its intranet is accessed using the HTTPS protocol.
Explain why networking protocols are required. [2]
 - (d) Alice is transmitting a document to Bob as an email attachment. Alice has signed the document with a digital signature.
 - (i) Explain the purpose of a digital signature. [2]
 - (ii) Explain how a digital signature is generated and sent to the receiver. [3]
 - (e) Eve has intercepted Alice's transmission and has modified the document.
Explain how Bob will know that Alice's original document has been intercepted and modified. [4]